GMR (Generalized Moment Retrieval) 实验全量汇总大表

实验评估日期: 2026-07-22 ~ 2026-07-23 | 评估数据集: Soccer-GMR (Validation) | 包含所有实验轨迹与单项指标最强结果高亮标记

最高 MAP (全场)	最高 MAP (EATR骨干)	最高 AUROC	最高 REJ-F1@0.4	最高 G-MIOU@3 (全场)	最高 G-MIOU@3 (EATR骨干)	最高 MR+@5
9.16%	**9.09%**	**73.31%**	**48.12%**	**36.40%**	**21.10%**	**4.17%**
Strict Moment HieA2M v2	EaTR + Quality (best-mAP)	EaTR + Counter	EaTR + Dual gate	Strict Moment HieA2M v2	EaTR + Dual gate	EaTR + Quality

1. EaTR 跨骨干迁移实验全量结果 (EaTR DGQC Transfer Experiments)

EaTR 方案变体	AUROC (%)	Rej-F1@0.4 (%)	mAP (best-joint) (%)	mAP (best-mAP) (%)	mR@5 (%)	mR+@5 (%)	G@1 (%)	G@3 (%)	对应跑的结果/Checkpoint路径
Plain EaTR (基线)	69.59	0.00	8.73	8.73	12.85	1.37	6.63	3.51	`artifacts/eatr_dgqc_transfer/seed2023/frozen_parent/eatr_plain_b128_best_map.pt`
Strict GMR (EaTR-GMR)	71.67	39.35	8.02	-	12.30	2.93	20.15	16.82	`artifacts/eatr_dgqc_transfer/seed2023/eatr_gmr_strict`
+ Quality (IoU排序头)	71.98	44.31	8.24	**9.09** ★	12.69	**4.17** ★	21.97	19.13	`artifacts/eatr_dgqc_transfer/seed2023/eatr_quality`
+ Dual gate (双路交互)	72.05	**48.12** ★	8.06	-	11.91	1.31	**24.04** ★	**21.10** ★	`artifacts/eatr_dgqc_transfer/seed2023/eatr_dual`
+ Counter (条件计数器)	**73.31** ★	29.79	6.16	8.66	9.46	0.93	15.28	12.15	`artifacts/eatr_dgqc_transfer/seed2023/eatr_counter`
完整 HieA2M-DGQC	72.65	22.66	7.08	8.72	9.89	1.61	13.21	9.56	`artifacts/eatr_dgqc_transfer/seed2023/eatr_hiea2m`

2. 两级判空与学习式选框 / 时域去重消融全量结果 (Two-Stage Selector & Dedup Ablation)

父模型基线：Strict Moment-HieA2M v2 (Direct Top-3 mAP = 6.48% / Stage 1 Geometry Top-3 mAP = 7.57%)

消融分类 / 实验分支	Direct Top-3 mAP (%)	消融/学习式 Top-3 mAP (%)	mAP 增益 (%)	mR@3 (%)	G@3 (%)	最优参数配置	平均选框数	对应结果路径 / 总结文件
Stage 1: Hard / DIoU NMS	7.57	7.92	+0.35	-	-	IoU threshold = 0.5	3.0	artifacts/validation_selector_ablation/seed2023/stage1_geometry/dedup_ablation_summary.json
Stage 1: Soft-NMS (Linear)	7.57	7.92	+0.35	-	-	Linear decay	3.0	
Stage 1: Complete-link Fusion	7.57	7.96	+0.39	-	-	IoU threshold = 0.5	3.0	
Stage 1: Linear Soft-NMS Top-5	8.67 (Top-5)	8.84 (Top-5)	+0.17	-	-	Top-5 预算	5.0	
Stage 4/5: **seed2023 主线** (posw=2.0)	6.48	7.05 ★	+0.57 ★	9.21	9.45	K=3, lambda=2.0	3.0	artifacts/validation_selector_ablation/seed2023/stage4_5_selection/learned_selector_ablation_summary.json
Stage 4/5: **seed2023-posw1** (posw=1.0)	6.48	7.05 ★	+0.57 ★	9.21	9.45	K=3, lambda=2.0	3.0	artifacts/validation_selector_ablation/seed2023_posw1/stage4_5_selection/learned_selector_ablation_summary.json
Stage 4/5: **seed2023-posw4** (posw=4.0)	6.48	7.05 ★	+0.57 ★	9.21	9.45	K=3, lambda=2.0	3.0	artifacts/validation_selector_ablation/seed2023_posw4/stage4_5_selection/learned_selector_ablation_summary.json
Stage 4/5: **seed2024 稳定性验证**	6.48	6.79	+0.31	8.85	9.20	K=3, lambda=2.0	3.0	artifacts/validation_selector_ablation/seed2024/stage4_5_selection/learned_selector_ablation_summary.json
Stage 4/5: **seed2025 稳定性验证**	6.48	6.87	+0.39	8.92	9.30	K=3, lambda=1.0	3.0	artifacts/validation_selector_ablation/seed2025/stage4_5_selection/learned_selector_ablation_summary.json
5 组 Selector 实验平均	**6.48**	**6.96**	**+0.48**	**+1.19 (提升)**	**+0.11 (提升)**	**-**	**3.0**	**所有 5 组均获得一致正向提升 (+0.31 ~ +0.57)**
Stage 4/5: **Soft Count 软计数**	6.48	6.72	+0.24	10.69 (mR@5)	10.15	prior_weight=0.0	~5.94	提升 Recall，但输出框偏多导致 mAP 略降
Stage 5: **软计数 + 谨慎边界融合**	6.48	6.72	+0.24	10.69 (mR@5)	10.15	未产生额外触发	~5.94	指标与软计数完全一致，暂不进入主方案

3. 跨骨干基线与组件变体全量结果 (Backbone Reproduction & Variant Matrix)

模型骨干 (Backbone)	组件配置 (Variant)	AUROC (%)	Rej-F1@0.4 (%)	mAP (%)	mR@5 (%)	G@1 (%)	G@3 (%)	协议/说明	对应跑的结果路径
Moment-DETR	**Strict HieA2M v2**	-	-	9.16 ★	-	-	36.40 ★	Strict (主方案父模型)	artifacts/formal_strict/moment_detr/seed2023/md_hiea2m_b128_rerun_from_best_v2
	Strict GMR v2	-	-	8.02	-	-	16.82	Strict 配对基线	artifacts/formal_strict/moment_detr/seed2023/md_gmr_b128_rerun_from_best_v2
	GMR Release Anchor	-	-	8.50	-	-	-	官方发布 Anchor	artifacts/anchors/moment_detr_gmr_release
QD-DETR	Plain (Staged)	-	-	7.32	-	-	2.60	Staged 基线	artifacts/formal/qd_detr/seed2023/qd_detr
	QD-GMR (Formal)	-	-	-	-	-	-	Legacy loss (ep 71)	artifacts/formal/qd_detr/seed2023/qd_detr_gmr
	QD Quality (Formal)	-	-	-	-	-	-	Legacy loss (ep 54)	artifacts/formal/qd_detr/seed2023/qd_quality
	QD Dual (Formal)	-	-	-	-	-	-	Legacy loss (ep 59)	artifacts/formal/qd_detr/seed2023/qd_dual
	QD Counter (Formal)	72.31	26.15	7.19	13.49 ★	-	9.35	计数精准度 36.34%	artifacts/formal/qd_detr/seed2023/qd_counter
	QD HieA2M (Formal)	-	-	-	-	-	-	Legacy loss (ep 63)	artifacts/formal/qd_detr/seed2023/qd_hiea2m
	Plain bsz128 Restart	53.32	42.02	0.07	0.00	-	17.22	定位训练塌陷	artifacts/canonical_b128_restart/qd_detr/seed2023/qd_detr
CG-DETR	Plain (Formal)	-	-	-	-	-	-	Formal 旧协议 (ep 132)	artifacts/formal/cg_detr/seed2023/cg_detr
	Plain bsz128 Restart	43.34	0.00	0.01	0.02	-	1.55	严重训练塌陷	artifacts/canonical_b128_restart/cg_detr/seed2023/cg_detr
EaTR (Plain)	Plain (Staged)	-	-	7.92	-	-	2.99	Staged 旧基线	artifacts/formal/eatr/seed2023/eatr
	Plain bsz128 Restart	69.59	0.00	8.73	12.85	6.63	3.51	定位正常, 拒答失效	artifacts/canonical_b128_restart/eatr/seed2023/eatr